CV

Sigrid Trägenap

Computational neuroscientist, Frankfurt Institute for Advanced Studies and Goethe University Frankfurt. Web: sigridtraegenap.github.io · E-Mail: traegenap AT fias.uni-frankfurt.de · ORCID: 0000-0002-8845-452X

Education

11/2018 – 2026	**PhD student**, International Max Planck Research School for Neural Circuits, Frankfurt. Thesis *Emergence of reliable sensory representations in cortical development*, submitted 01/2026. Supervisor: Matthias Kaschube
06/2020 – 11/2024	Member, **Center for Multiscale Modelling**, Frankfurt
10/2016 – 10/2018	**MSc Biophysics**, Goethe University Frankfurt. Thesis: *The role of network interactions in refining neural response properties in visual cortex*. Supervisor: Matthias Kaschube. Final grade 1.0
10/2013 – 11/2016	**BSc Biophysics**, Goethe University Frankfurt. Thesis: *Development of coding properties in mammalian visual cortex*. Supervisor: Matthias Kaschube. Final grade 1.3
2009 – 2013	**Landesschule Pforta**, Department of Natural Science. Abitur, final grade 1.0

Research experience

01/2019 – 03/2019	Rotating PhD student with **Pascal Fries**, Ernst Strüngmann Institute, Frankfurt. Analysis of electrophysiological recordings, role of α-band activity
10/2017 – 10/2018	Student assistant with **Matthias Kaschube**, Frankfurt Institute for Advanced Studies. Analysis of spontaneous activity in visual cortex
2016 – 2017	Student assistant with **Gilles Laurent**, Max Planck Institute for Brain Research, Frankfurt. Sepia project
2011 – 2012	Intern, Martin Luther University Halle-Wittenberg, group of Prof. Binder, macromolecular chemistry

Teaching

2021 – 2024	**Theoretical Neuroscience I & II**. Teaching assistant: guest lectures, coordination of exercise sheets and grading, content preparation
2018 – 2026	**Computational Neuroscience Seminar**. Teaching assistant: paper selection, introduction to scientific writing, feedback on oral and written presentations
2017 – 2018	**Machine Learning I**. Teaching assistant
2015 – 2016	**Mathematics for Physicists I**, **Mathematics for Scientists II**. Undergraduate teaching assistant: weekly tutorials, grading

I have also supervised several Bachelor's and Master's theses on machine learning, image processing, data analysis and network simulation.

Selected talks

- 10th European Visual Cortex and Cognition Meeting, Cortona, 2026 (poster selected for short talk)
- Rhine-Main Neuroscience Meeting, Oberwesel, 2026 (invited)
- Giersch Conference on Digital Twins, 2023
- Computational Neuroscience Next Generation Symposium, 2023

- Vision Sciences Society Meeting, St. Pete Beach, 2023
- German Neuroscience Society Meeting, Göttingen, 2023 (Young Investigator Talk)
- Society for Neuroscience Annual Meeting, San Diego, 2022 (nano-symposium)
- Ferret Brain Meeting, San Diego, 2022
- Bernstein Conference, Berlin, 2022
- Annual Computational Neuroscience Meeting, Barcelona, 2019

Selected posters

- Bernstein Conference, 2026
- HWK Workshop "Cortex Structure and Function in Mouse, Monkey and Man", Delmenhorst, 2026
- Cosyne, Lisbon, 2026, 2024 and 2019; Montreal 2023; virtual 2021
- Bernstein Conference, Frankfurt 2025 and 2024; virtual 2020; Berlin 2018; Göttingen 2017
- Meeting on Spontaneous Activity in Brain Development: Alicante 2024 and 2022, Leiden 2018
- 9th European Vision and Cognition Meeting, Il Poggio, 2024
- Biennial Meeting of the Rhine-Main Neuroscience Network, 2018

Summer schools

- Okinawa Computational Neuroscience Course (OCNC), 2022

Awards and scholarships

2023	Travel Award, Organization for Computational Neuroscience
2022	Travel Award, Bernstein Network
2022	**Poster Award (1st prize)**, 2nd Meeting on Spontaneous Activity in Brain Development
2019	Travel Award, Organization for Computational Neuroscience
2019	Cosyne Presenters Travel Award
2018	**Poster Award (3rd prize)**, 5th Biennial Meeting of the Rhine-Main Neuroscience Network
2017 – 2018	SMARTSTART training programme in computational neuroscience, Bernstein Network and Volkswagen Foundation
2014 – 2018	Scholarship, German National Academic Foundation (Studienstiftung des deutschen Volkes)
2013 – 2014	Deutschlandstipendium
2013	DPG Abiturientenpreis, German Physical Society
2013	Honours of eLeMeNTe e.V., Saxony-Anhalt

Service and outreach

2024	Co-organiser, PhD symposium at FIAS following the Bernstein Conference
2019 – present	Co-organiser of the German Neuroscience Olympiad final in Frankfurt: outreach and press contact, grading coordination, computational tutorial
2018 – 2022	Student representative, Frankfurt Institute for Advanced Studies
2013 – 2018	Student representative (Fachschaft Physik), Goethe University Frankfurt: faculty council, examination committee, lecture evaluations

Skills

- **Languages**: German (native), English (C2), Latin
- **Programming**: Python (advanced), MATLAB, LaTeX